IBA Student App Development, User Testing and Evaluation

Prototype link: https://samanwaseem.github.io/iba-app/

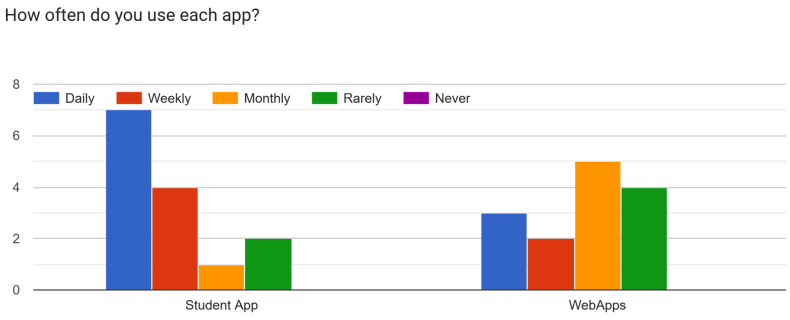
By Norman Door Fixers (Saman Waseem 29258, Hadeeqa al Islam 27173)

Objective

We aim to make the Student App less redundant and something students would find worth using every day by adding the surveyed most frequently used but hard to access and navigate WebApp services (transport, exams, gate pass, career services). This will be done using multiple heuristics encouraging daily use.

Problem

The findings from our Survey 1: (https://docs.google.com/forms/d/e/1FAIpQLSekt_DshDHyAtN-SrEQlIV0NdjBwm0Rvzu5CeCGGF0IVtB4Mw/viewform?usp=dialog) show that the Student App is **used more frequently** but for attendance (indicating lack of flexibility and redundance) while the WebApps have **better functionality** but are difficult to navigate or access quickly through the phone for frequent use.



Scope

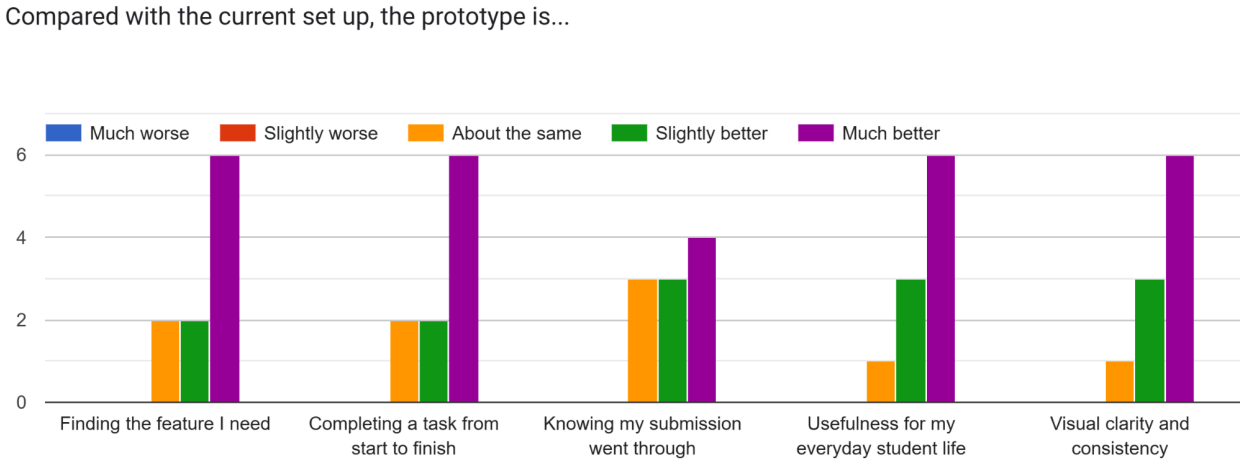
The work focuses on integrating the WebApps services into the Student App, instead of redesigning them. Admit cards, transcripts and internship forms were named as the more used services in Survey 1. Shuttle timings, point bus sign-up, gate passes and recommendation letters are routine student tasks that currently sit outside the Student App, and Survey 2 confirmed the value of adding the shuttle features.

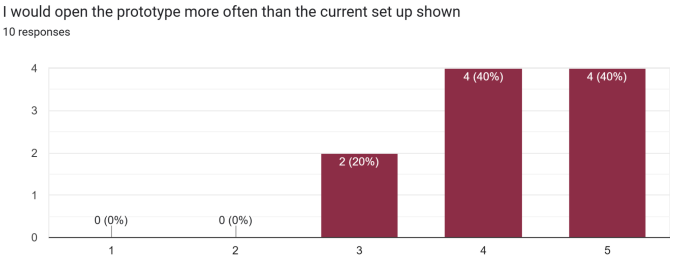
Target Users

Our users are IBA students who use the Student App on their phones and WebApps on the web throughout the semester.

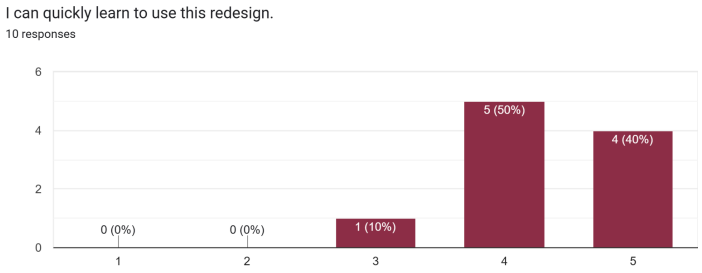
Formative Evaluation of the Prototype: Survey Questions and Responses

Survey 2 (changes were implemented according to insights from this): https://docs.google.com/forms/d/e/1FAIpQLSfbMtXm2m1WZa6BXw-izMBMV8ytd17WTq3J5aMpksjuHh9sHw/viewform?usp=dialog

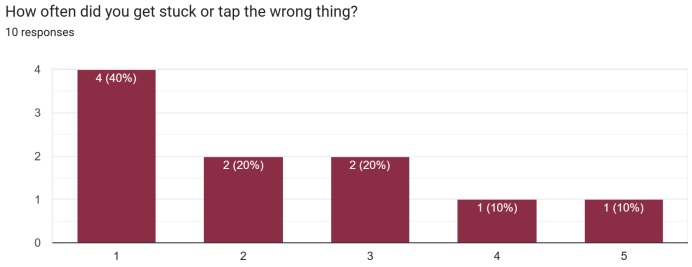




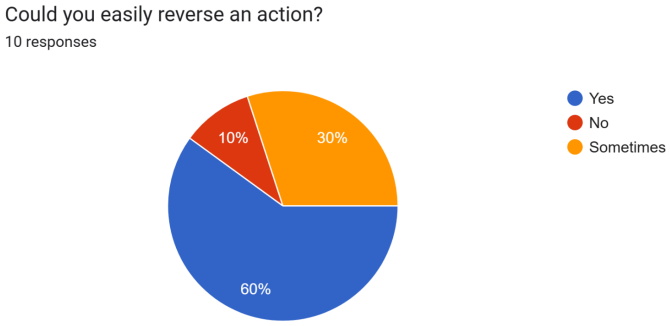
(Strongly Disagree on the left, Strongly Agree the right)

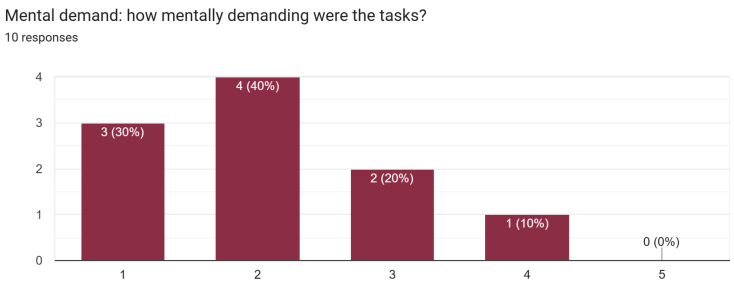


(Strongly Disagree on the left, Strongly Agree the right)

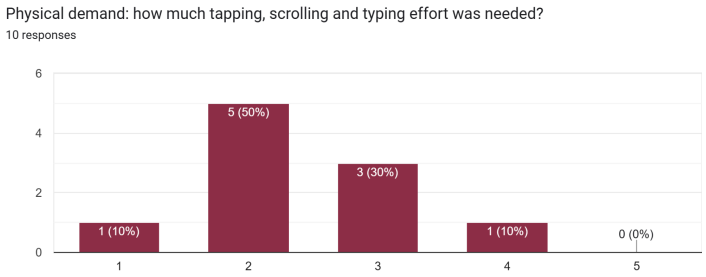


(Never on the left, Always on the right)

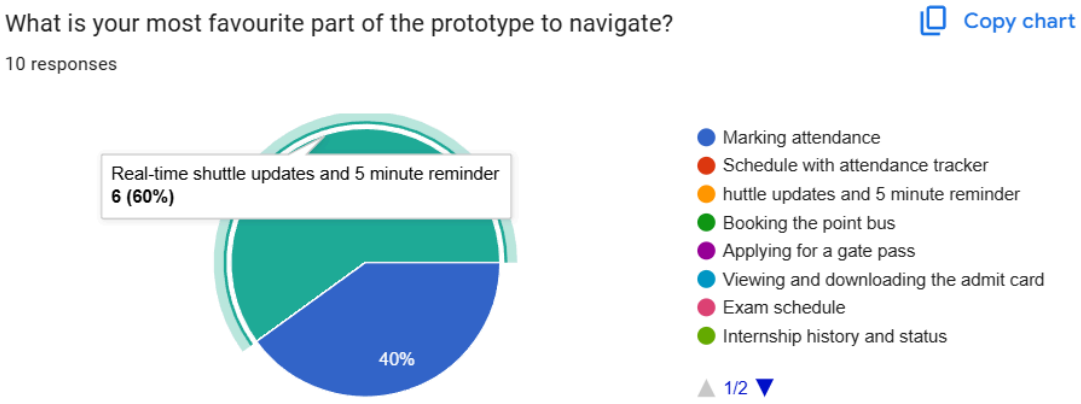




(Very low to Very high)



(Very low to Very high)



Why is it your favourite?

Don't need to search emails for shuttle timings

there's no need to find the email in outlook and then figuring about the shuttle timings

Well, info for the shuttle and point are all scattered through email / transport website, so its actually a hassle to look for this separately. If the student app had this readily available, id actually have a reason to use it other than to mark attendance. The admit card is also a plus for this exact reason.

i had to put in no effort. very easy on my mind.

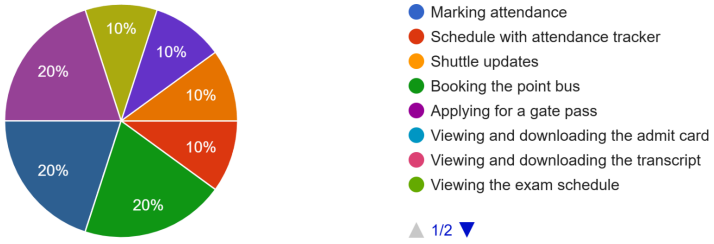
Daily use

Useful daily because right now it’s hard to know when the next shuttle will be we have to check the website and calculate

It is easy to use

What was your least favourite part of the prototype to navigate?

10 responses



Why is it your least favourite?

Cannot edit submission

The attendance marked screen is a little hard to read because it blends in with the background. The popup when you successfully mark your attendance should have its own solid background to make it stand out against the homepage of the app.

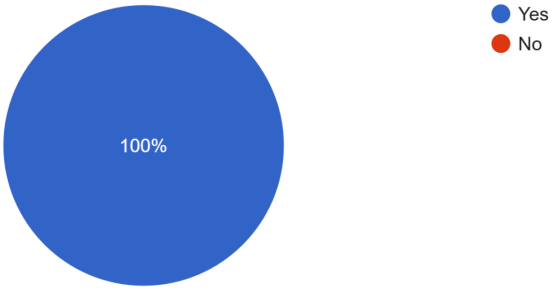
it is in itself a needy task.

Not frequently used

A little bit confusing at first glance

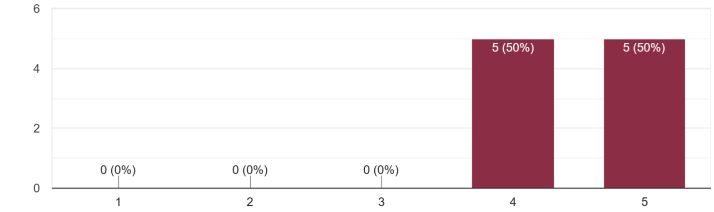
Is this redesign something you would find worth opening daily?

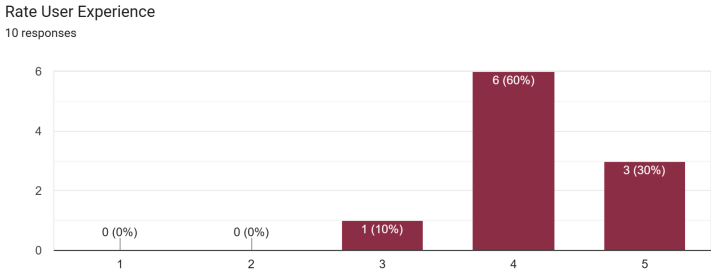
10 responses



Rate the Interface

10 responses





Did anything make it hard to use, such as text size, colours or touch targets? 10 responses

No Not really nope Not really, it was pretty intuitive, Nope, some text colours made them camouflage with the background making them a bit hard to read., no, - , NA

What is missing that you would want in a combined IBA app? 10 responses

I would want assignment notifications (opening and deadlines) to show up too

a whole profile like a cv so we can reach out to companies with it.

Im not sure if youve already taken this into account or not, but whenever there are some changes to the transport timings, such as for exams or other events, its a hassle to try and find the latest email with the new temporary timings. It would be nice if your app could show the new temporary timings as well according to any emails from the transport office.

N/A

maybe assignment reminders

Reminders

n/a

Qualitative themes

- Less searching for timings: A participant wrote that there is "no need to find the email in outlook" to work out shuttle timings, and another that they "don't need to search emails for shuttle timings". A participant noted that it is currently hard to know when the next shuttle will be.
- A reason to open the app beyond attendance: A participant said that having the shuttle and point information in the app would give them a reason to use it other than marking attendance.

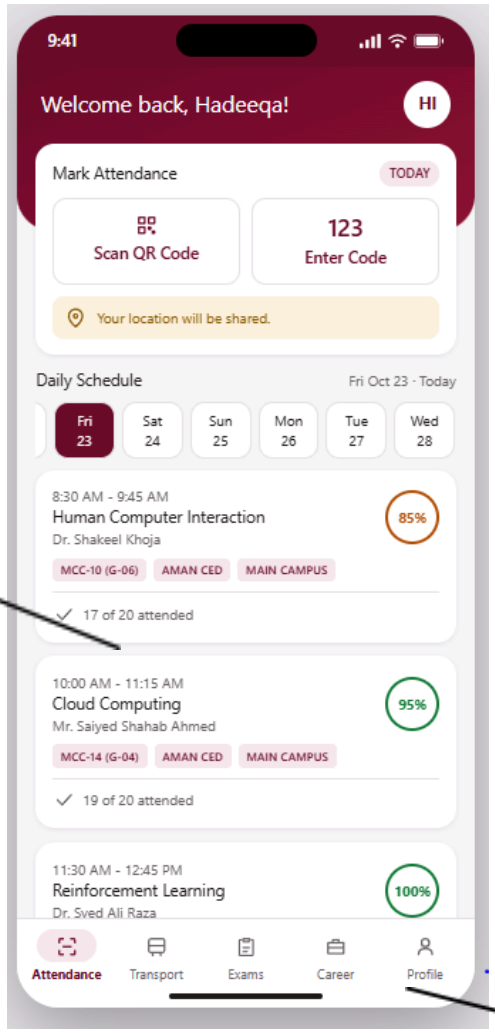
- Low effort: A participant wrote that they "had to put in no effort" and found it "very easy on my mind".
- Readability: One participant reported that some text colours blended with the background, and another found the attendance-marked screen hard to read for the same reason. Both were addressed in iterations.
- Submission control: A participant named the inability to edit a submission as the least favourite part.
- Missing features: reminders, assignment notifications, a CV-style profile and temporary shuttle timings shown in the app (now implemented).

Comparison of Old Setup vs. New Prototype in Terms of HCI Principles

Aesthetic and Minimalist Design - Nielsen
Discoverability - Norman

The schedule and listed courses used to occur thrice in three different tabs which was redundant.

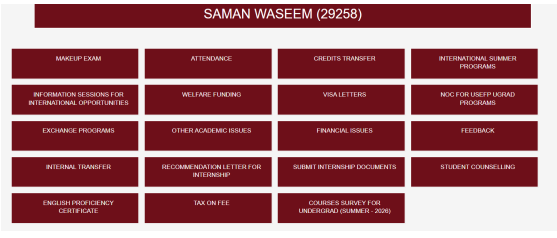
This combines attendance tracking.



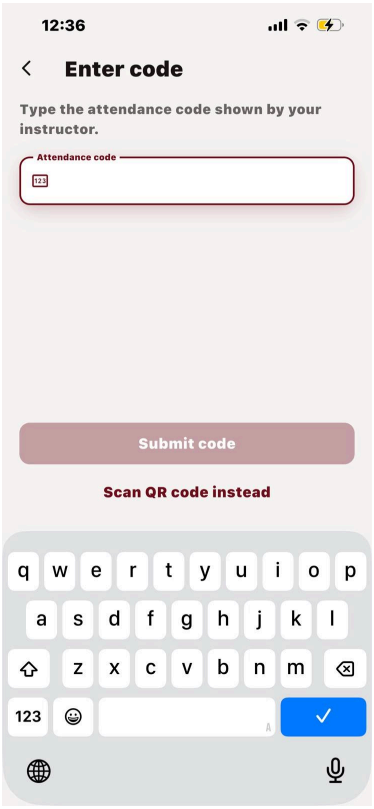


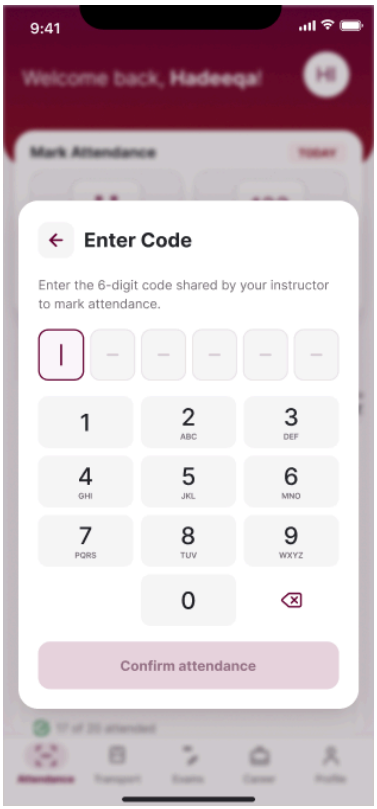
Reduce short-term memory load - Shneiderman (cognitive load reduction)

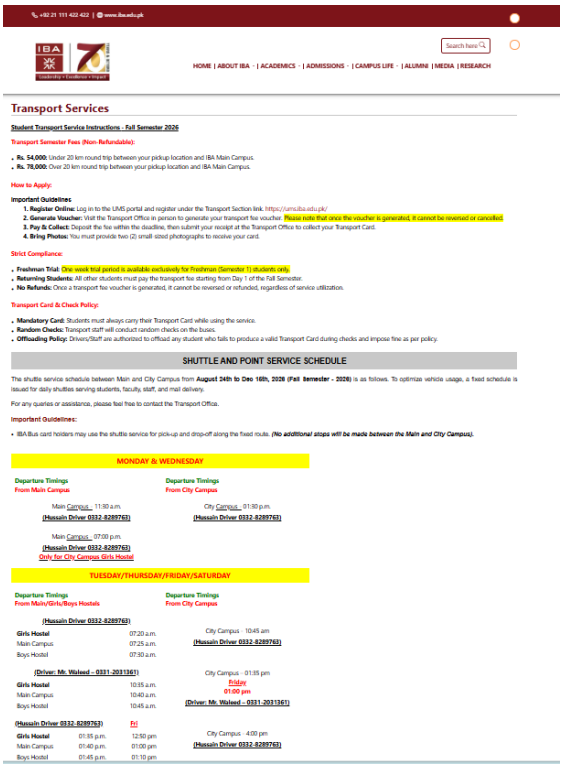
Recognition rather than recall - Nielsen

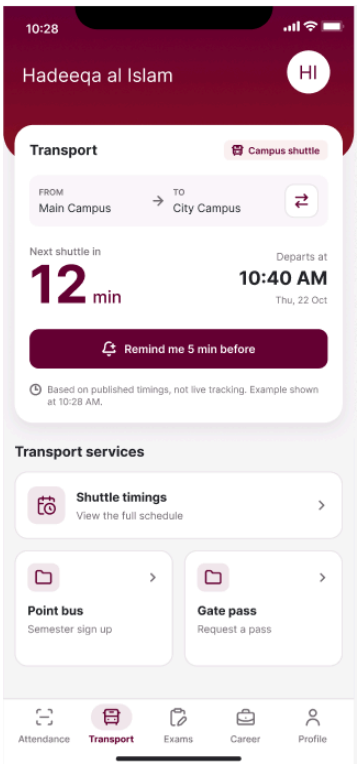


Error prevention - Nielsen



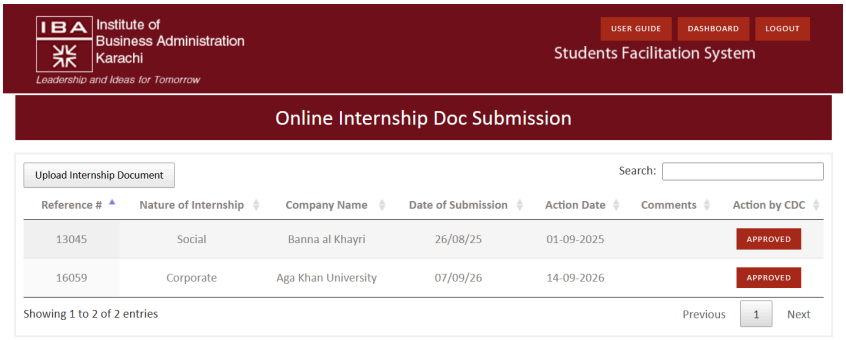


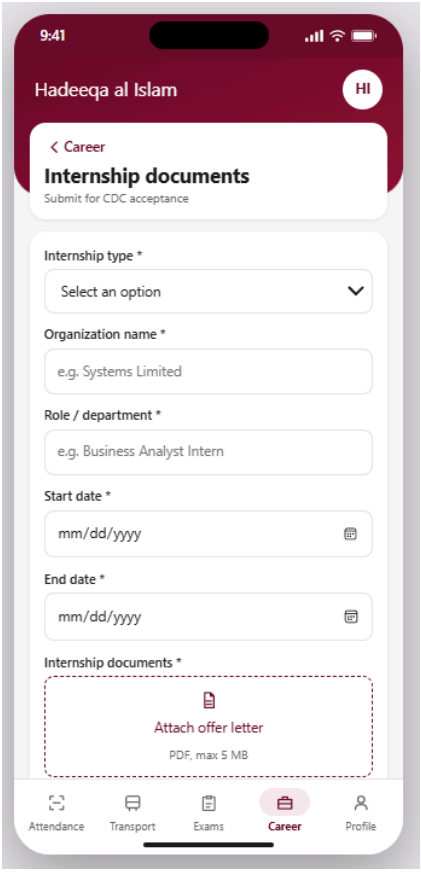




Discoverability - Norman

Appreciated by multiple users in the survey since shuttle timings and updates are currently used every single day by many yet difficult to locate when urgently needed.





Reversibility (internship document submissions are editable until the submission deadline) - Shneiderman

Hence discoverability - Norman

Universal Usability - Shneiderman

Visibility - Norman

More HCI Principles Applied

We justify design decisions with Nielsen's usability heuristics, Norman's principles of visibility, feedback, constraints and consistency, and Shneiderman's golden rules.

- Visibility of system status and feedback: the shuttle card shows the minutes until the next departure, the reminder button changes to Reminder set, step indicators show progress through the point bus and gate pass flows, and every submission ends on a confirmation screen with a summary.
- Error prevention: the code keypad accepts six digits and keeps Confirm disabled until all six are entered, forms check the CNIC, phone number, email and date order before moving on, and confirm or cancel dialogs sit in front of final actions.
- User control and freedom: every sub-screen has a back link named after its parent screen, dialogs can be cancelled, and internship submissions can be edited until the deadline.
- Consistency and standards: five tabs with the same icons and labels on every screen, a shared header and white title card, the same primary button and status chips in every tab. Career was restyled to match the other tabs.

- Recognition rather than recall: tasks are grouped under familiar names (Transport, Exams, Career), dropdowns list the valid options, and entered details are summarised before the final step.

Aesthetic and minimalist design: one schedule instead of three views, one primary action per screen, and removal of an explanatory note on the Transport card that repeated what the countdown already shows.

Conceptual Design

Information architecture

The prototype has a login screen and five top-level tabs in a persistent bottom bar.

- Attendance: mark attendance (scan a QR code or enter a six-digit code) and a daily schedule with an attendance percentage for each course.
- Transport: next shuttle with a direction swap and reminder, shuttle timings (temporary and regular), point bus sign-up and gate pass requests.
- Exams: admit card, ongoing transcript, and an exam schedule with Midterms and Finals, term and year filters.
- Career: internship documents, internship history and details, recommendation letter requests and letter request tracking.
- Profile: personal details, notification preferences, privacy and security, help and support, and sign out.

Task models and user flows

The five tasks below were used in the evaluation. Each is described as a hierarchical task analysis with the steps in order.

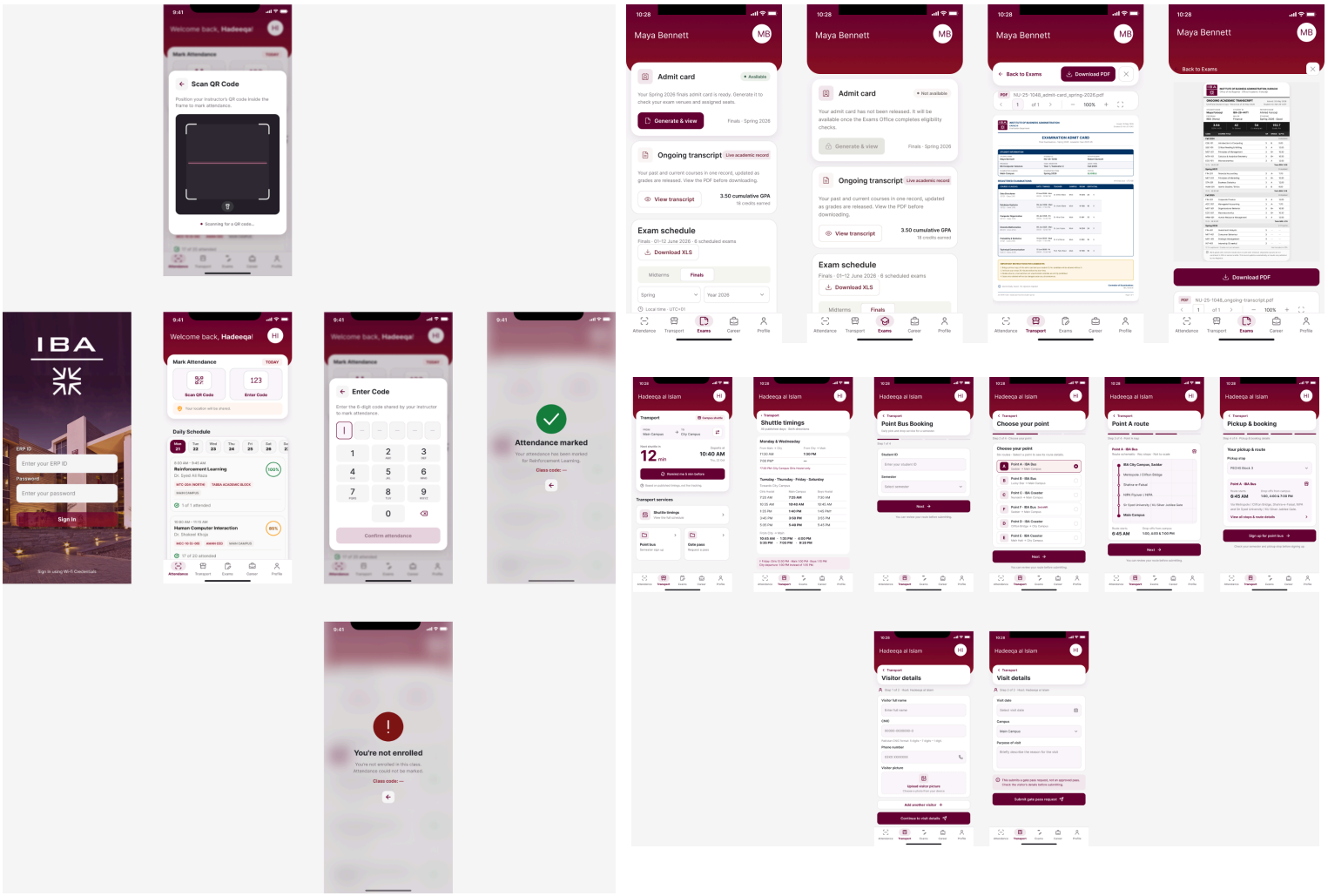
- Task 1, mark attendance: open the Attendance tab; choose Scan QR code or Enter code; scan or type the six-digit code; confirm; read the result (attendance marked or not enrolled).
- Task 2, sign up for the point bus: open Transport; choose Point bus; enter student ID and semester; choose a point from A to F; review its route; choose a pickup stop; confirm in the dialog; read the confirmation.
- Task 3, request a gate pass: open Transport; choose Gate pass; enter the visitor's name, CNIC and phone, with an optional picture or another visitor; continue; enter the visit date, campus and purpose; submit and confirm; read the confirmation.
- Task 4, open the admit card: open Exams; choose Generate and view; check the venue and seat; optionally download the PDF.

- Task 5, request a recommendation letter: open Career; choose Recommendation letter; enter purpose, recipient, program, deadline and delivery method; review; send and confirm; find the request under Letter requests.

Evaluation support: For testing, the attendance code 123456 succeeds and 000000 shows failure.

Wireframes

We designed wireframes in Figma for the login, attendance, transport, exams and career screens before building. The prototype follows them closely, including the four-step point bus flow and the two-step gate pass flow.



Prototype Development

The prototype is a single-page mobile web app written in HTML, CSS and JavaScript. It is laid out for a 390 by 844 pixel phone screen and hosted on Github so that participants can open it

from a link on their own phones. It uses simulated data for a fictional student, so no real student information is stored. Every button, dropdown, form and dialog works, and screens scroll.

We sketched it on paper then developed it on Figma both manually and through the built in AI agent. Manual edits were required to achieve the exact look and functionality we had in mind. The AI assistant also helped produce an interactive prototype.

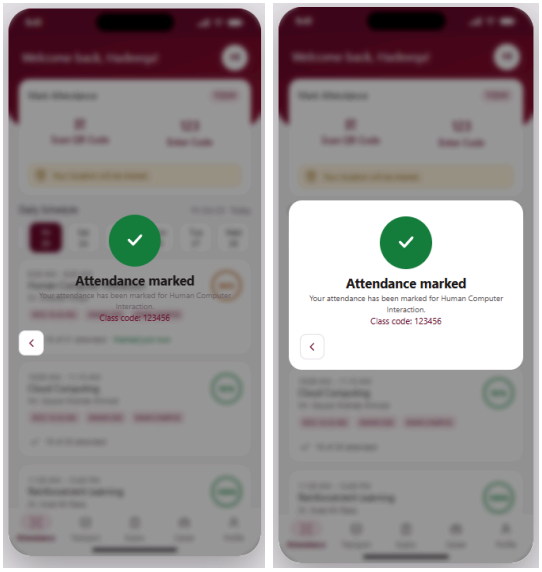
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Iterative Refinement

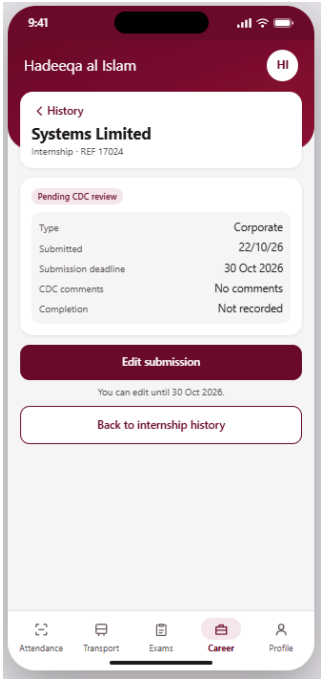
We refined the prototype in three kinds of rounds: team testing of the first build, comparison with the wireframes, and Survey 2 feedback.

Phase 1: Changes made after insights from Survey 2

- Better readability



- Editing internship submissions as reversal of actions (only while the deadline has not passed)



- More confirmation dialogue boxes for every action. Bigger ones for what needs more acknowledgement and smaller ribbons similarly according to priority.

Phase 2: Changes made through survey responses received after Phase 1's updated link

- Easier to go back to the previous state by tapping anywhere on the grey area surrounding a dialogue box or opened screen. This also allowed 'Cancel' for easy reversibility
- More aesthetic log in page background
- Temporary shuttle timings added (mid-term week, summer semester, etc) from Muhammad Siddiqui Athar (27875)'s insight: *"I'm not sure if you've already taken this into account or not, but whenever there are some changes to the transport timings, such as for exams or other events, it's a hassle to try and find the latest email with the new temporary timings. It would be nice if your app could show the new temporary timings as well according to any emails from the transport office."*

Phase 3: Final Review

- Added dynamic animations on the header upon view
- Added animations to the log in page

Feedback not yet implemented and why

- Assignment and quiz deadlines: lecturers do not publish deadlines on any IBA system, so the app would depend on students typing them in. A reminder feature inside the schedule risks clutter, so we list it as future work.
- CV-style profile: outside our scope of moving frequently used WebApps tasks into the app.
- Recommendation letters and internship documents: the reasons given were low frequency and the nature of the task, not the interface.